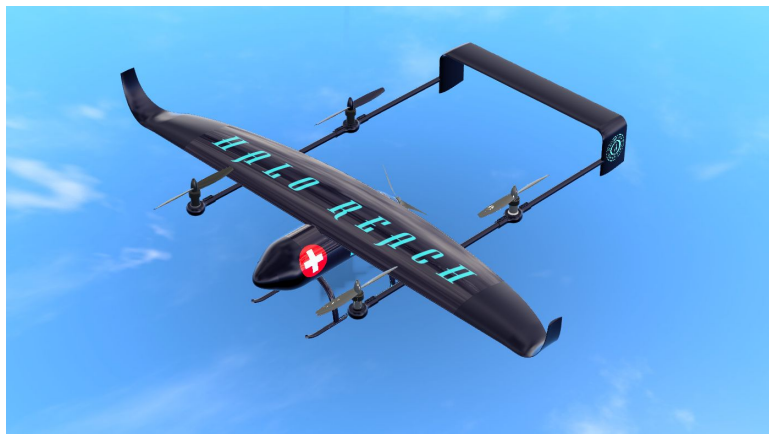

HALO REACH: Structural Analysis and Design of the Modular-Add Ons

UAS for Disaster Relief - AE3 2024



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Abstract

Every year, powerful earthquakes hit parts of Turkey and Syria, leading to the loss of thousands of lives. The proposed UAV will transport relief and medical aid to these countries with damaged infrastructure and in dangerous weather conditions. This report explores the design process of the UAV modular add-on, focusing on 2 particular objectives. Firstly, a user friendly and robust design for the modular add-on is required. Secondly, the modular add-on must be able to incorporate payloads for agricultural and infrastructure management needs, to branch into other markets. A particular focus on the structural analysis and buildup of the design iterations was made in this report. Each design iteration was tested by running static structural simulations, along with a validation of the obtained results. The results of the structural simulations showed that the design had low stress concentrations, which proved a robust and safe design. The clear thought process behind choosing the payload attachment mechanism allowed for a user-friendly design, which could easily accommodate other payloads for different markets. In future iterations, a more detailed structural analysis would be preferred, with the simulation of each separate component under certain loading. Furthermore, it would be beneficial to prototype this mechanism to observe its performance, allowing for better improvements.

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Nomenclature

L Length of the plate

1 Introduction

1.1 Brief and constraints

This project is based on the 41st Annual Student Design Competition, with the title ‘Multi-Mission Modular UAS for Disaster Relief’. The main aims of the modular add-on team include designing the add-on and the interaction point between the add-on and the UAV. Furthermore, the stability of the add-on must be ensured, from a structural, aerodynamics and aeromechanical point of view. In particular, this report will focus mainly on the structural analysis of the modular add-on. To briefly introduce the mission and the use of this UAS, there are 2 main mission profiles: the supply delivery mission and the long-endurance mission. The supply delivery mission contains medicine, vaccinations, medical aid kits or rations. The long-endurance mission is expected to contain communication items to allow for relay purposes. Further details about these two missions can be found in table 2.

Table 2. Key details for the 2 missions

Mission	Endurance (hours)	Payload weight (kg)	Payload dimensions
Supply delivery	No prescribed endurance	50	Given volume of 0.05m^3
Long-endurance	10	20	Cube of volume 0.05m^3

The most important requirements applied to the modular add-on subsystem were the following and presented in the VFS Brief[1]:

1. “The vehicle modifications will take place between missions on the deck of a ship.”
2. “The maximum total weight of the add-on(s) for each mission configuration must be no greater than 15% of the GTOW.”
3. “The modular add-on(s) must be swapped out and in by 2 people in less than 10 minutes.”
4. “The interface of the modular add-on(s) and the vehicle must be well-defined.”

Furthermore, the modular add-on team was further constrained to a 7kg maximum weight due to the mass distribution of the UAV.

1.2 Other markets

Focusing on the modular aspect of this UAS project, the add-on was designed to enable the UAV for many applications in other markets. It was decided that both the payloads would be cubes of volume 0.05m^3 . This would mean they must have sides of 36.9cm (a prescribed density of $1000\text{kg}/\text{m}^3$ was given for the supply delivery mission). Having collaborated with the project coordination team to finalise other markets for this UAV, the following markets were decided upon. [2] [3]

- Large scale delivery services.
- General land monitoring.
- Agriculture and pesticide spraying.
- Infrastructure management.

Focus on other markets was quite useful whilst designing the modular add-on system and this will be spoken about in section 4.4.

1.3 Payload position

To understand the design, a brief summary on the chosen placement of the add-on is required. It was decided that the payload would be stored outside the fuselage as it allows easier access and has a greater scope for the integration of cameras, pesticide sprays and other containers. An external payload implies that only the payload would be changed when this UAV is used for different markets, as opposed to redesigning the entire fuselage. This characteristic makes this UAV more appealing and adds onto the modularity feature.

Secondly, there were multiple ways to externally store the payload, including hanging it underneath the fuselage, or hanging from a ramp in the back. Though these ideas had great potential, the add-on ended up acting as a bluff body in the flow field, significantly increasing the drag. Thus, it was chosen to cut a slot into the fuselage and insert extra load-carrying structures to support the weight of the payload. The payload will be stored inside a container which will be further discussed in section 4. Figure 2 shows the slot cut into the fuselage and the payload container would be a cube with sides 37.1cm. The fuselage would continue to wrap around the payload container from the sides too, but these were removed for a better view of the add-on. Note that this figure is not representative of the final CAD for this project, but it is for demonstrative purposes at this stage in the project.



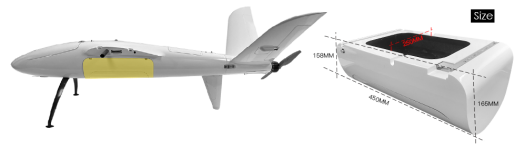
Figure 2. Payload slot in the UAV.

2 Payload Attachment

2.1 Initial ideas



(a) Multicopter KWT-X6L-15 [4]



(b) YANGDA Sky Whale [5]

Figure 3. UAV examples from research

Initially, a mind-mapping approach was used to develop many ideas for attaching the payload container to the external of the fuselage. Current UAVs in the market tend to attach their payload to the external of the fuselage by simply manufacturing the payload into the fuselage mainframe. Furthermore, most payloads are usually a specific camera or electrical equipment, and the mainframe is often specialised for this exact purpose, as shown in figure 3a.

A good example of a UAV with an external payload is the YANGDA Sky Whale, shown in figure 3b, which made use of a quick detaching assembly, for a max payload of 10kg. [5] This was an excellent case and helped to define some desired characteristics to finalise attachment options. All the considered attachment options can be found in Appendix A. The main characteristics were:

1. Ease of use.
2. Structural compatibility to current fuselage and weight addition.
3. Aerodynamic effects and increase in drag.
4. Simplicity.
5. Power consumption.
6. Versatility/ modularity.
7. Reliability.

These characteristics were based on research of commonly used attachment options and the given brief. At this point in the project, a non-numerical method was adopted to finalise some options. This was deemed acceptable as further numerical investigation was carried out about the final 3 options to gain a better understanding of their performance.

2.2 Final 3 options

Once the final characteristics were chosen, a Pugh matrix was used to narrow down the options. A specific weighting was put on each characteristics based on a deep understanding of the missions, and the different attachment options were rated on a scale of 1-5, where 5 is the highest. For further detail on this selection process, see Appendix B. In conclusion, the final 3 options can be seen in figure 4. To choose one final option, the structural analysis of the 3 payload attachments was carried out, highlighting the key benefits and drawbacks of each option.



Figure 4. Final 3 options for payload attachment

2.2.1 Initial model setup and details

The initial structural analysis was carried out to get a rough estimate for the shear scale of deflection and stresses to be expected in each scenario. The fuselage connection points were modelled as a flat plate of dimensions 300mm x 1300mm x 5mm. The flat plate was made out of CFRP, as the material of the fuselage was unknown at this point. The material for the payload container was also undecided, and was set to CFRP, as it was quite likely to be a composite. A 1.5 safety factor was applied to the weight of the payloads in all of these cases, to account for any uncertainties and ensure reliability. The main focus will be the points of maximum stress as this shows where the majority of the load is being carried, and the parts needing the most attention in the next iteration.

2.2.2 Sliding rails

The idea of sliding rails was inspired by Boston Dynamics' Spot payload mounting mechanism.[9] This idea would include sliding payloads onto the rails and using a locking mechanism to prevent the payload from moving during flight. In this very simple analysis, this locking mechanism was not included and the rails were assumed to have taken all the load of the payload. Mounting rails and slotted channels were used as inspiration in choosing the initial dimensions for the sliding rails, but then modified to make the inner section longer to better carry a 50kg payload in a stable manner. [10] An immediate problem with the weight of the rails was noticed, and further iterations would be required to minimise the mass of the rails decide on a lightweight material. To structurally analyse the rails, the payload was represented as a point mass at the centre of the payload. The results can be seen in figures 5.

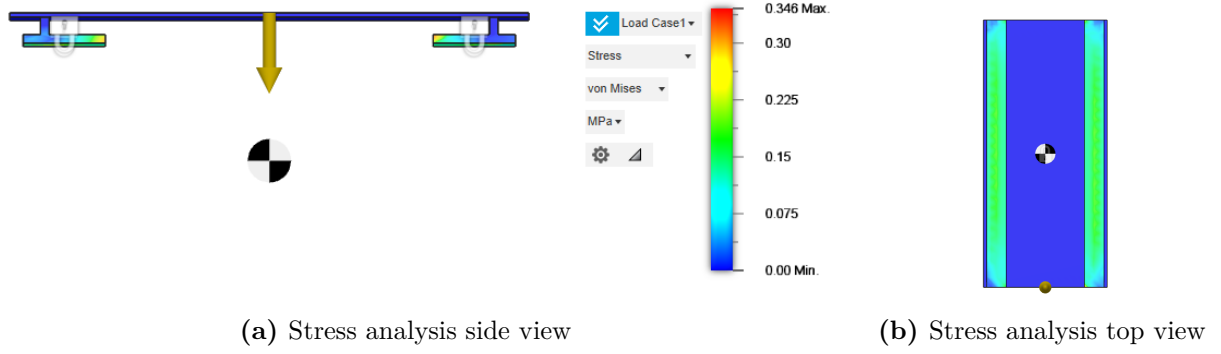


Figure 5. Initial stress analysis of sliding rails

The stress levels are generally quite low, but are concentrated at certain sections. As seen, the highest stress points tend to be the inner corner of the rails holding onto the payload, thus a redesign of that part would be beneficial. The stress on this particular area will be worsened with the extra weight of the locking mechanism and the effects of vibrations during flight. This would need to be heavily considered in future iterations, as vibrations in the payload can cause significant damage to the fuselage and UAV control dynamics.

2.2.3 Electromagnetic fasteners

Electromagnetic fasteners were a fascinating idea, mainly due to their low manual effort to use and low price. To model this idea, it was decided to store the electromagnets inside the fuselage and have the payload attached to the external surface of the fuselage. This protects the electromagnets from any external debris. As Fusion 360 is unable to model magnetic behaviour, it was decided to observe how the plate and the payload container would deform if simply attached to the fuselage, with a fixed boundary condition (forming the worst case scenario). A fixed boundary condition means that the top of the payload that is in contact with the plate is fully fixed and no displacements are allowed at that surface. An electromagnet was found that could carry the 75kg payload and this was used to model the correct dimensions and weight of the electromagnets. [11] Because this configuration leads to the payload being the least supported, the payload was not represented as a point mass, to see any potential displacements on the actual payload box. Figure 6 shows the stress distribution.

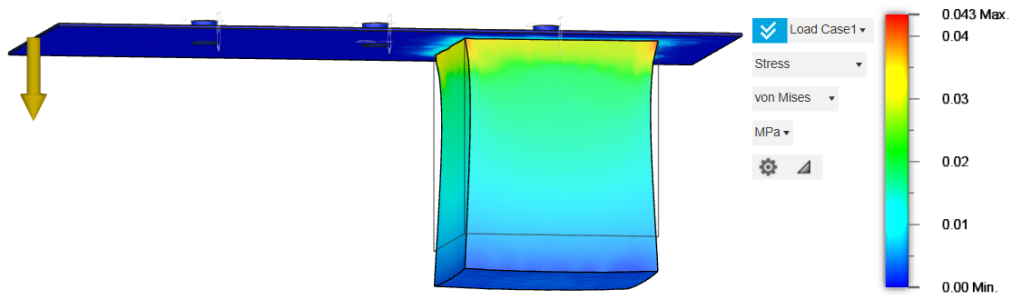


Figure 6. Initial stress analysis of using electromagnetic fasteners

2.2.4 Mechanical clamps

The design for the mechanical clamps were inspired by the design of forklifts. Figure 7 shows the stress analysis of this design. The payload was represented as a point load in between the top and bottom plate, and it can be seen to bend the entire structure. There is a high stress concentration in the upper part of the middle joint, and this would need to be designed well in the next iteration if continued.

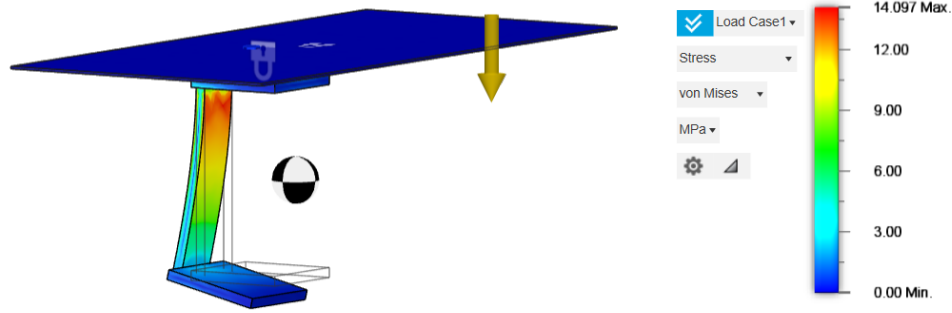


Figure 7. Initial stress analysis of using mechanical clamps

2.3 Payload attachment discussion

To choose a final design, table 3 shows the maximum stresses and an initial weight estimate of the 3 designs. The initial weight estimates were calculated by looking at similar products on the market (such as rails, electromagnets, etc.), and adding all the weights for the entire mechanism as a whole. For example, the weight estimation for the electromagnetic fasteners includes the weight for the electromagnet, the mounting mechanism and the extra batteries to power the electromagnet.

Table 3. Maximum stresses and weight estimations for the final 3 attachment options

Attachment option	Max stress (MPa)	Weight estimation (kg)
Sliding rails	0.346	3.2
Electromagnetic fasteners	0.043	7.7
Mechanical clamps	14.097	5.4

Evaluating the results in table 3, the sliding rails have a really low mass and a low maximum stress. However, there were many concerns with locking the payloads in place so as to not produce extra vibrations during the missions. Furthermore, loading and unloading the sliding rails would be quite difficult as it may be quite hard to load a 50kg payload onto sliding rails.

The electromagnets are a great option due to their low mass, however it requires 2kg batteries to power it for the 10 hour long-endurance mission, which significantly increases the weight of the entire mechanism. Electromagnetic interference with the communication signals and equipment was another concern with this idea. It would require the fuselage to protect the electromagnets in a specialised material which reduces the chances of electromagnetic interference. It was decided that this was far too high of a risk, especially as the communications are a really important aspect of a disaster relief drone, and thus this design was not chosen.

The mechanical clamps have the highest stress point in comparison to the other designs. Mainly, this is concentrated at the middle clamping plate, and this option would require quite detailed redesign and careful structural analysis. It was noticed that a rack and pinion system could be used to attach the upper and lower plates together, as opposed to the original singular plate. This is greatly beneficial as it allows the controlled lowering or raising when the payload is being swapped. The other designs require a new mechanism to stop the payload being dropped upon release, which may require a lot of manual effort to load and unload. This point is a very important benefit of this design, as it ensures that we accomplish point 3 in the brief, as mentioned in section 1.1. Thus, due to this very point, the mechanical clamp was chosen to be the final decision as it had the most potential. A careful redesign is required, to ensure the design is fail safe and this will be explained from section 3 onwards.

3 Structural buildup and analysis

To make the second iteration of the mechanical clamp, it was decided to remove the top plate so the top of the payload container would be pushed up against the fuselage. A new gear and pinion rack was introduced to act as the connection point where the rod would be attached to the back heavy frame, the design of which can be found in Rosenthal A.'s report.[12] The plate would be attached to the racks using aerospace grade steel bolts, to allow for easy removal and replacement of the plate. [12] It was agreed that the motor for the gears can be controlled by the pilot, meaning that an automated lowering and raising of the system can be achieved. Iteration 2 can be seen in figure 8. The main objectives in the structural buildup and analysis were:

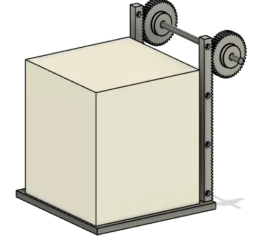


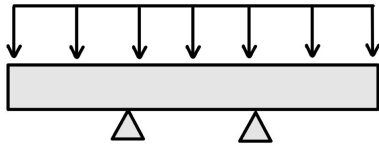
Figure 8. Clamp mechanism iteration 2

1. Decide on rail placement along the length of the bottom plate.
2. Create a lightweight but strong design for the bottom plate.
3. Prove and validate that this mechanism is able to securely handle the weight of the payload.

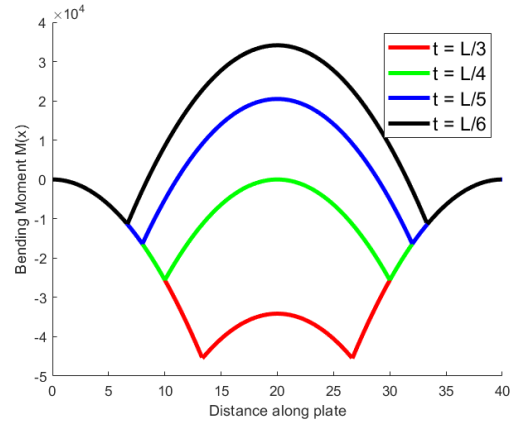
3.1 Rails Placement

The first calculation to consider was the placement of the rails with respect to the plate to correctly distribute the load. This was done analytically by approximating the system to a simple beam under a distributed load with 2 fixed supports (acting as the rails). This is shown in figure 9a.

Using the simple beam theory, the bending moment distribution can be written as a function of the rails placements, and this is shown on figure 9b.[13] When the rails are placed at $L/4$ (where L refers to the length of the plate) from either end, this results in the best bending moment distribution. The bending moment is 0 at the middle and it also has the lowest 'Maximum Bending moment' when compared to all the other rail placements. By placing the rails at the $\frac{1}{4}$ and $\frac{3}{4}$ points, the load of the payload is well distributed between the 2 rails, leading to the most robust and stable rail placement.



(a) Rail placement approximation



(b) Moment Distribution along the length of the plate.

Figure 9. Clamp mechanism rail placement

3.2 Bottom plate design

The entire modular add-on subsystem is constrained by a 7kg limit. The bottom plate is a crucial part of the add-on, as it needs to correctly support the payload. The material of the bottom plate was

decided to be Epoxy/HS Carbon fibre, the reasoning for which is discussed in Nakhla A.'s report.[14] A 2cm solid plate of this material would weigh over 5kg, which is far too large and needs to be redesigned. The following sections will explain how a lighter plate was designed.

3.2.1 Generative Design

Generative Design was used to iterate through multiple design solutions that meets certain constraints given by the user. In this case, a generative design study was used to create a minimum mass plate that could support the 50kg payload. The starting shape was a 2cm thick plate, from which the software was constrained to only remove excess material. An original structural simulation of a 2cm thick plate was carried out and showed encouraging results. However, it had a large safety factor of 12 in the simulation, indicating a lot of excess mass. The most promising solution generated was figure 10a, from which figure 10b was created. The solution from Generative Design was 1.2kg, which was significantly lower than the starting shape. Its thickness was 12mm at its thickest points, and 8mm at its thinnest. Figure 10b was a plate of 10mm and uniform thickness, to ensure that the payload would not slip down the plate whilst unloading.

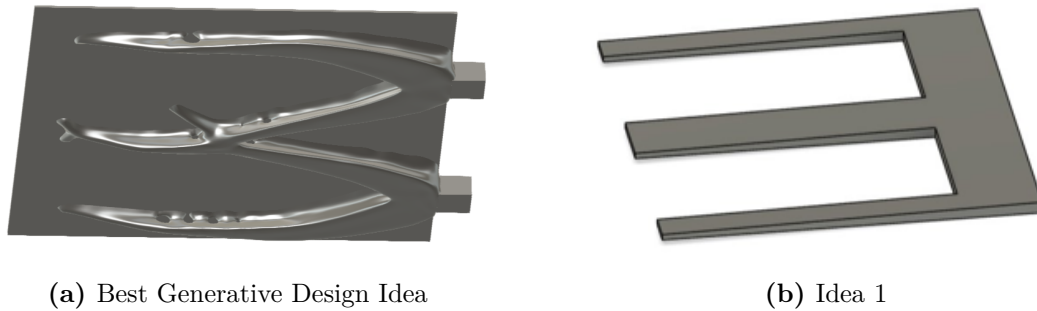


Figure 10. Bottom plate design iteration 1

3.2.2 Mesh Convergence Investigation

A mesh convergence must be carried out to determine the best mesh size to use for the consequent structural simulations. It allows the user to ensure that the results are independent of the mesh size, and includes refining the mesh size until the results are consistent. For this investigation, the maximum von Mises stress was studied. Table 4 shows the results of a mesh convergence study in Fusion 360 for this setup. The mesh size for the first 3 iterations were model-based size, meaning that the mesh is a certain percentage of the model size. The last 3 iterations were using an adaptive mesh, which enforces a finer mesh at joints and high gradient-regions. [15] An adaptive mesh allows for great accuracy, but in a fraction of the time, thus better suited for the quick simulations required here.

Table 4. Mesh convergence investigation

Mesh size	Max stress (MPa)
10%	28.263
5%	27.4066
3%	104.773
Adaptive Mesh - Low	153.492
Adaptive Mesh - Medium	161.578
Adaptive Mesh - High	164.239

The maximum stress begins to plateau in value as you refine the mesh. Having said this, an adaptive mesh at medium level will be used for the rest of the analysis, in the interest of computational power and processing time.

3.2.3 Results from iteration 1

Figure 11a shows the stress distribution across the entire mechanism under the 50kg load (a 1.5 safety factor has been applied to every analysis). Figure 11b shows the corresponding displacement.

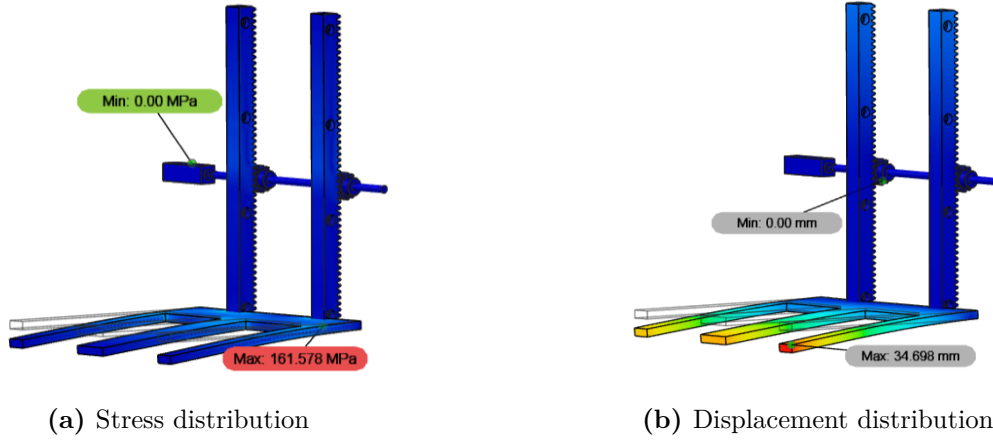


Figure 11. Structural analysis of the bottom plate iteration 1

Figure 11 shows a maximum stress of 161.578MPa, and a maximum displacement of 34.698mm. The loads are mainly carried by the bolts holding the plate and the rack together and lead to a considerable displacement at the free end of the plate. As the bolts were made from aerospace grade steel, they were expected to carry the given load. Further investigation was carried out to relieve the stress on the bolts and to reduce the displacements on the plate.

3.2.4 Final iteration

To reduce stress on the bolts, it was decided to add stringers that attach the bottom plate to the top of the fuselage heavy frames. Multiple ideas were discussed to reduce the stress on the bolts, and this will be discussed in Nakhla A.'s report, along with the selection and placement of the specific stringers and spring reels.[14] Stringers and spring reels were chosen as they would ensure tension in the stringers whilst raising and lowering the payload. The stringers were attached as shown in figure 12a. The particular advantage of this setup was that the add-on would be attached to the UAV fuselage through 2 heavy frames, instead of just 1, as was the case in the previous design iteration.

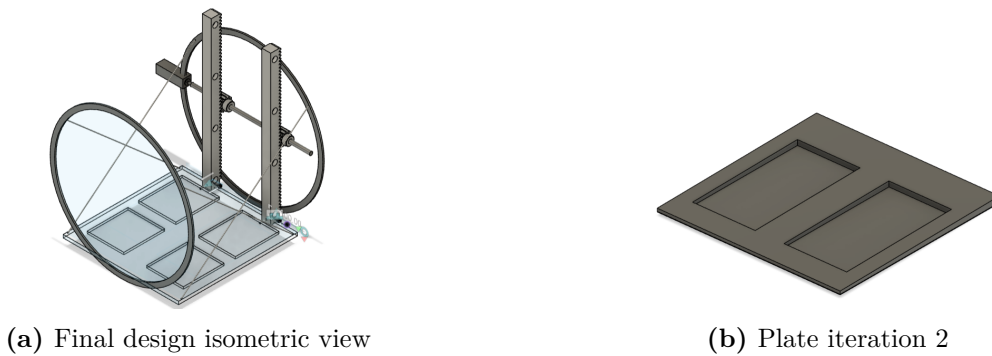


Figure 12. Further images showing the final design and iteration 2 of the plate

The plate would need multiple new iterations to adjust for the addition of the stringers. Firstly, the bottom of the plate was completely covered up, to prevent recirculation when the payload container is not stored in the clamp. The second change was made by adding a strut joining the three struts together was placed at the free end. This enables the 3 struts to work together, resulting in a more uniform handling of the load. Furthermore, it also allows the middle strut to be supported by the

stringers, reinforcing a more stable design, which can be seen in figure 12b. Upon analysing such a plate, it was realised that more structural reinforcements were required in that direction, eventually leading to the design in figure 12a. Each design iteration was tested separately in a static structural simulation, and this design had the lowest weight and lowest maximum stress. At its thickest point, it was 10mm and 3mm at its thinnest point, leading to a total weight of 1.23kg. Multiple iterations were made in between the first and the last iteration, which can be found in Appendix C. These designs were created with the particular uses of the UAV in mind, especially when used in other markets. It was crucial that the plate would still allow for the addition of a camera in the middle of the plate, and some pesticide nozzle slots.

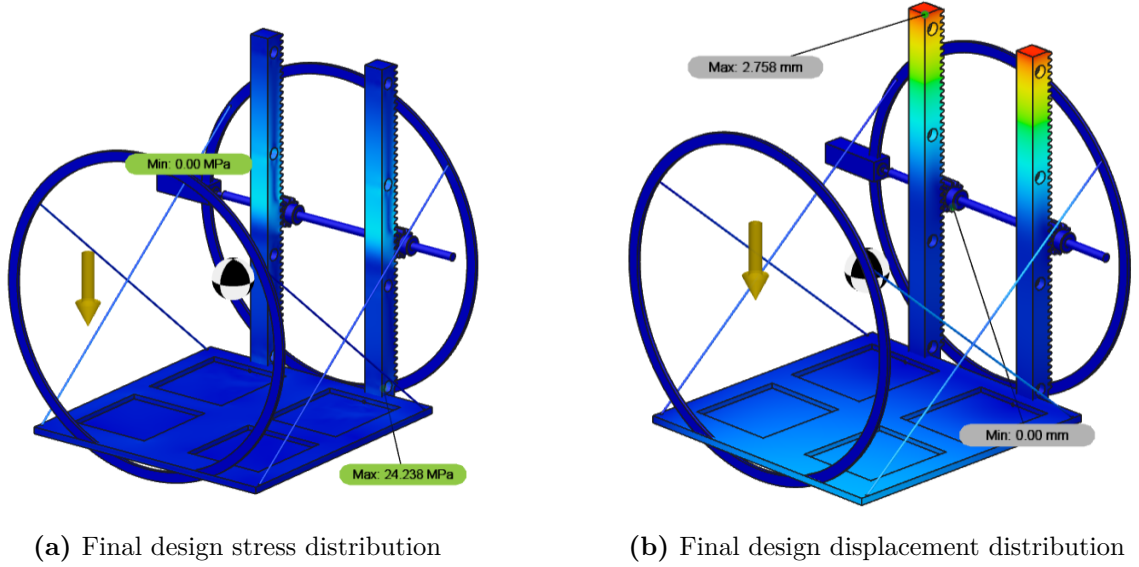


Figure 13. Final Design structural analysis

Figure 13a shows the stress distribution of the final design. It can be seen that the stress is being taken by the stringers as well, reducing the stress concentration at the bolts. The maximum stress is lowered to 24.238MPa, still concentrated at the bolts. Furthermore, the maximum displacement was reduced significantly to 2.758mm and was no longer at any edge on the plate, but instead on the rails. This was deemed acceptable due to its small magnitude, and is mainly focused at the top of the rails, which is not in use when the payload is stowed up during flight.

3.2.5 Results validation

All previous analysis was carried out in AUTODESK Fusion. However, it is known that Abaqus CAE is a better software for more accurate stress distribution results.[16] Abaqus CAE is a dedicated finite-element software, which can handle complicated geometry and nonlinear analysis, often leading to more precise results. However, it was quite difficult to represent this mechanism in Abaqus CAE, due to its complexity with different materials, pinned connections and moving mechanisms. Thus, Fusion's faster run time, ease of use and static simulation environments were preferred over the course of this 5-week project. To obtain a more accurate result of the stress distribution on the plate, an analysis on the plate individually was carried out. All the bolts, stringers and payload weight were resembled as separate forces and boundary conditions applied to the plate without the external parts. This analysis was not expected to line up perfectly with the results from Fusion, but instead, act as a means of validation for the original results. This study is unable to accurately simulate the behaviour of the bolts, the stringers and the contact with the rails, meaning the plate is under a slightly different set of loading. However, this study will help to reasonably understand the behaviour of the plate with more precision.

The bolts were represented by enforcing a fixed boundary condition in the threads of the plate, and the stringers were represented by enforcing a pinned boundary condition at their locations. Furthermore, a fine hex mesh was chosen to correctly capture the behavior around the bolts and other areas of large

gradient changes. Figure 14 shows the final result of this analysis. The maximum stress is on the outer edges of the bolt threads, which aligns well with the Fusion analysis. The maximum stress in Abaqus CAE is 13MPa, which is in the same order of magnitude as the Fusion analysis, with a maximum stress of 24MPa, thus correctly validating the earlier analysis. Figure 14 also shows some clear deformation over the plate, which was not seen in the Fusion analysis. This is an obvious discrepancy between the 2 models and this result was to be expected as the stringers were modelled as simple pinned connections instead of load-bearing structures. However, in reality we expect the stringers to carry some of the load, and reducing displacements on the plate.

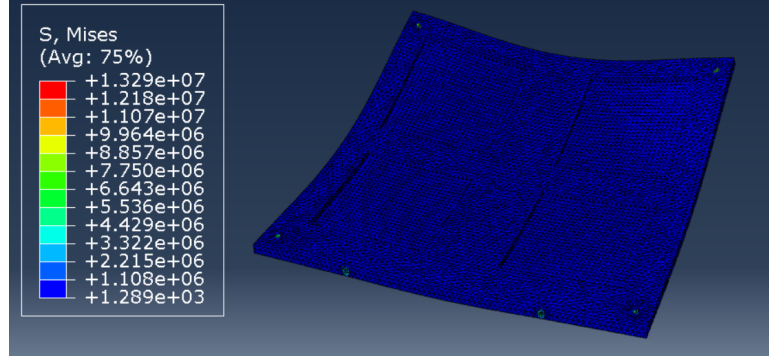


Figure 14. Deformed stress distribution in Abaqus CAE

4 Payload container design

As part of the add-on, the payload container needed to be designed. To obtain the modularity objective of this project, the payload container would be changed depending on the use of the UAV and be able to serve other purposes too.

4.1 Payload container material

4.1.1 Important qualities

This UAV will be used in quite dangerous environments, one that will have debris hitting it, in extreme weather conditions and earthquake-prone environments, leading to large vibrations. [17] By carefully considering this environment, the following qualities were considered quite important for the container material to have:

1. Specific strength
2. Fatigue strength
3. Specific stiffness
4. Fracture toughness
5. Thermal expansion coefficient
6. Density
7. Cost

The specific strength, specific stiffness and fracture toughness were important quantities that were telling of the material's strength. Although the payload will be stored in a slot within the fuselage, it is important that it can also survive if any debris hits against the fuselage. The fatigue strength is an important quality because it is expected that this UAV will be used multiple times, leading to a high number of stress cycles applied on the UAV. Thermal expansion coefficient ensures that the material is less likely to change size in varying temperatures. Given the environment that this UAV will be used in, a low thermal expansion coefficient is required. The density and cost are important as we want the lightest and cheapest material possible that fits the other criterion too. Lastly, the material manufacturability is important, as it must be easy to manufacture many payload containers for different uses upon demand. This will enable the payload material to be modular and easy to integrate into the original plans.

4.2 Commonly used materials

As there was not a numerical way to pick a material given the above qualities to maximise, common materials used to payloads and fuselages were analysed, due to their similar environments of use. Common aerospace materials often include Aluminium, CFRP, Titanium alloys and PEEK.[18] A range of materials were included, from metals to composites to evaluate a broad range of characteristics. Table 5 shows the evaluation of the common materials against each other. The values for this table have been obtained from the Aerospace 3 database on CES Granta Edupack.[19] The best material for each category was highlighted green.

Table 5. Comparing common aerospace materials for the payload container

	Al (7075-T6)	CFRP	Ti-6Al-4V	PEEK
Specific strength (kNm/kg)	164 - 189	700	203	68.7-84
Fatigue strength (MPa)	152 - 168	500-700	634-688	39-47
Specific stiffness (MNm/kg)	25.2669	43.750	25.4 - 25.9	3.0385
Fracture toughness (Mpa.m ^{0.5})	26.6 - 26.8	21.875	104 - 114	2.73 - 4.3
Thermal expansion coeff. (10 ⁶)	24.1	-0.1 to 1.7	8.7 - 9.1	47
Density (kg/m ³) (10 ³)	2.77- 2.83	1.6	4.43	1.3 - 1.32
Cost (GBP/kg)	3.05 - 3.5	21.5	17.2 - 20.9	42.9 - 44.9

From table 5, CFRP was the best material due to it's high strength, durability and high thermal stability, but was found to be quite expensive.

Having formed a baseline for some of the common aerospace materials, further materials were investigated to find a lighter and cheaper option than CFRP. There was a particular focus on plastics and composites as they are typically lighter. This investigation into other materials can be found in Appendix D. However, the outcome of the further investigation only confirmed the decision of choosing CFRP due to its unbeatable specific strength, fatigue strength and fracture toughness. Plastics were only marginally lighter but significantly weaker and less durable, and most composites were not suitable for this purpose.

4.3 Final details

Container wall thickness Given the 7kg constraint for the entire modular-add on system, there was a dedicated 1.45kg for the payload container, which gives a maximum thickness of 1mm for the container wall. Originally this seemed quite low and there were concerns about the strength of the payload container. However, carrying out more research into the use of CFRP in aerospace and container applications, it was decided that 1mm was an acceptable thickness for the container. As an example, shipping containers have steel walls that are between 1.6mm - 2mm thick. [20] Thus, using CFRP, which is nearly 10 times stronger than steel and of a similar thickness makes the design quite acceptable.

Container handles As the walls of the container are only 1mm thick, it was decided not to attach a concave handle. Instead, general cargo straps were used as they are lightweight and require little extra space. The cargo straps can be attached to the container using special fasteners made for composite materials. Extra reinforcements, such as additional CFRP layers near the fasteners may be required. [21] Accounting for the weight of the fasteners and lightweight cargo straps, this fits just within the 7kg constraint. [22]

4.4 Other markets

The bottom plate and the payload container should be modifiable for different uses and markets, especially as the chosen material for both is quite well-known and easy to manufacture with. The main concept of the modular add-on is that it is able to easily incorporate a system into the drone,

such as a thermal imaging camera, or a pesticide spraying mechanism, upon the request of any buyers. There will be mounting points at the bottom of the plate, allowing pesticide spray nozzles or a camera to be mounted to. For the pesticide-spraying mechanism, a pesticide tank, pump system, spray nozzles and corresponding sensors are required. With the given volume of 0.05m^3 , a maximum amount of 35L of pesticide could be stored, with the rest of the volume housing a pump system to pressurise the pesticide liquid.[23] A sealed cable mechanism at the top of the payload container would enable connection to the pesticide mechanism, allowing control of the valves and a power supply. Examples of these systems are shown in figure 15.

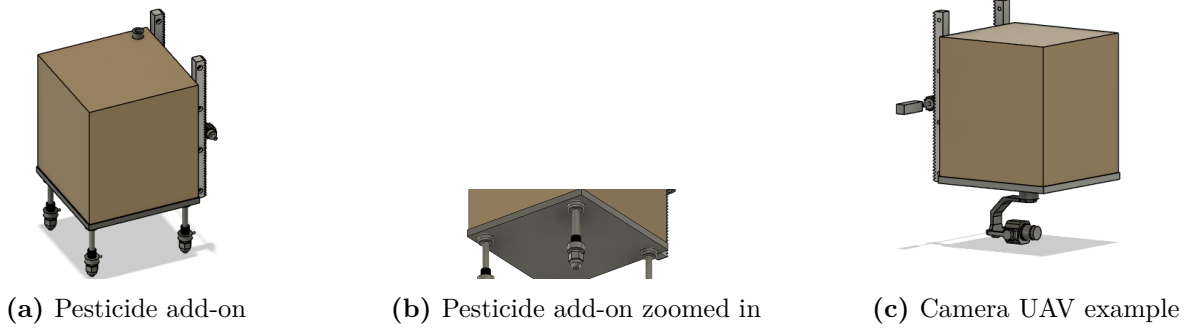


Figure 15. Other add-on examples

5 Conclusion and Improvements

To produce a stable and easy to use modular add-on, a variety of payload attachment types were analysed. By using a Pugh matrix and carrying out some initial structural analysis, the final attachment type was chosen to be the mechanical clamp. As this method allows an automated raising and lowering of the payload, it allows for a more user-friendly and quick to swap system. This decision enables the quick swapping of the payload by 2 people, meeting objective 3 for this project. Reflecting upon the selection method, it was noticed that the Pugh matrix relied on a subjective way to rate characteristics. Given additional time, a better selection process might have included the structural, aerodynamic and aeromechanical analysis of all original ideas before final selections. This would allow more innovative designs to be explored before choosing any final options.

With designing the mechanical clamp, there was a particular focus on the bottom plate to reduce the mass of the modular-add on subsystem. Using generative design tools, the original plate of mass 5kg was brought down to 1.23kg, with the incorporation of stringers to relieve stress on the plate and the rack bolts. Upon analysing this setup, the mechanism was found to be structurally sound, with very little stress concentrations. Iteration 1 saw a maximum stress of 161Mpa, which was eventually reduced to only 24MPa in the final iteration. This achieves the main aim of this report in that a structurally stable modular add-on has been designed. Realistically, computational simulations can take a long time to model accurately, thus it might be preferred to prototype this mechanism. This will allow for faster understanding and improvements. Furthermore, as the structural analysis of the bottom plate was validated using multiple simulations, the same should be carried out for other integral components. Lastly, further iterations should look into producing a bottom plate that has a more seamless integration with the UAV fuselage, leading to less drag and a better aerodynamic performance for the entire UAV. Some more features to investigate could include safety interlock mechanisms for increased safety on loading and durability during use.

The payload container was chosen based on a variety of different characteristics, the most important ones being fatigue strength, specific strength, density and thermal expansion coefficient. By focusing on these characteristics, an appropriate material was chosen for the payload container, indicating that the payload is well protected during the mission. This observation is very important as it is a critical part of the mission and the aims of this project. Next iterations might look into sandwich structures for the payload container, with specific thermal management methods (such as aerogels or polyurethane) for better thermal protection of the payload.

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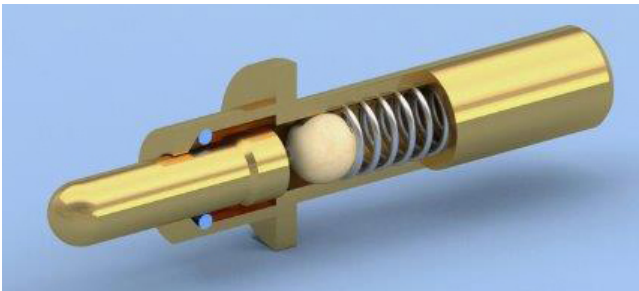
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Appendix

A Further attachment options

The following options were also considered:

1. Latch and lock systems.
2. Spring loaded pins.
3. Movable and adjustable straps.



(a) Spring loaded pins [24]



(b) Latch and lock systems [25]

Figure 16. Further attachment options



Figure 17. Adjustable straps [26]

Adjustable straps was then removed from the analysis due to its infeasibility to protect the payload and lack of structural capability with the fuselage.

B Selection process of attachment options

Table 6. Pugh matrix for payload attachment options

	Weighting	Rails	Latch and lock	EM	Spring loaded pins	Clamps
Ease of use	10	4	3	5	4	2
Compatability	8	3	3	2	3	3
Simplicity	7	4	4	5	5	4
Power consumption	5	5	5	2	5	5
Versatility	8	4	2	5	2	4
Reliability	7	5	3	3	3	5
Total	-	184	144	172	161	164

These characteristics were chosen based on a lot of discussion about the most important roles of this modular add-on. The characteristic, ‘Compatability’ refers to structural compatability and how well the mechanism can be attached to the UAV fuselage, as well as its own structural integrity. These were based on a mutual agreement within the modular add-on team, in an attempt to reduce any bias or subjectivity. The final 3 were the ones with the highest total score, highlighted in green in table 6.

C Structural analysis further iterations

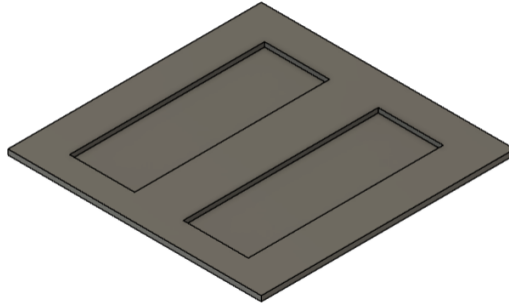


Figure 18. Iteration 3

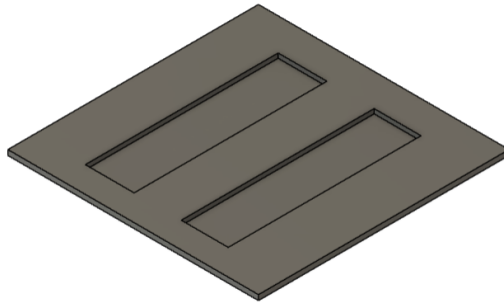


Figure 19. Iteration 4

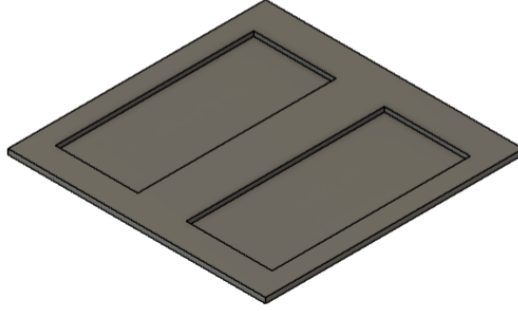


Figure 20. Iteration 5

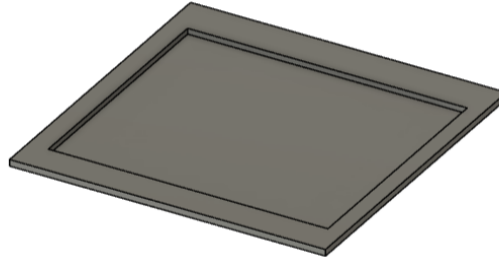


Figure 21. Iteration 6

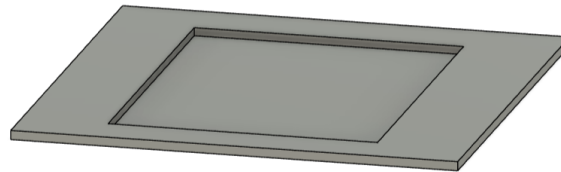


Figure 22. Iteration 7

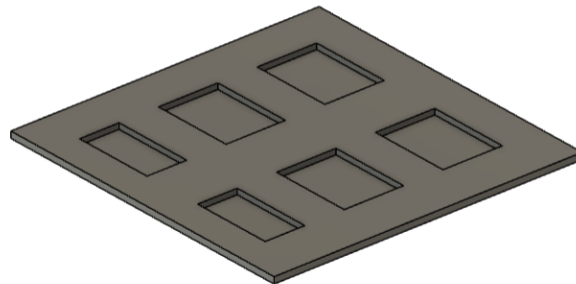


Figure 23. Iteration 8

D Other materials for the payload container

Considering all the different characteristics mentioned above, it was realised that the fracture toughness is the most important one, and thus a Ashby plot of fracture toughness against the density was made. Figure 24 was created using Ansys Granta CES software, and the composites with the highest fracture toughness are shown in the figure.

Nickel-Fe-Cr alloy and Nickel-titanium alloy were ignored as their densities were far too high. Checking the other 3 options, it should be noticed that they contain unidirectional fibres. This is when all the fibres are lined up in the same direction, leading to superior strength and fatigue resistant properties in the same direction, but really poor strength in any other directions. Unidirectional fibres are also

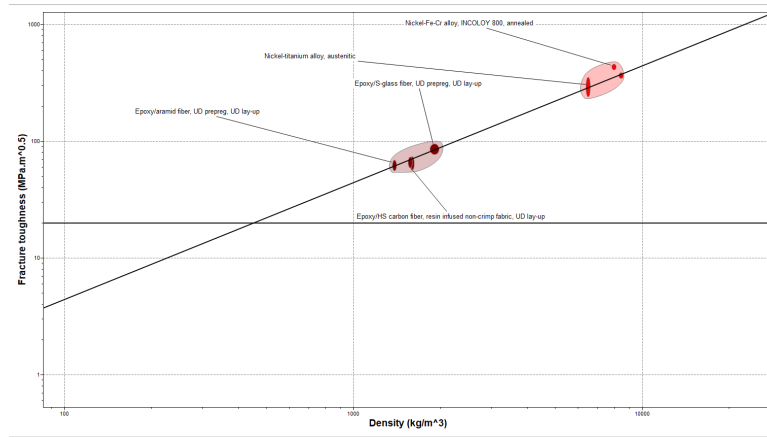


Figure 24. Ashby Plot showing the top composite materials for fracture toughness against density

quite likely to undergo delamination, which is when the fibre layers separate. Therefore, the focus was shifted towards plastic instead. Figure 25 shows a Ashby plot of fracture toughness against density for plastics, and it can be seen that PEEK (40% carbon fiber) and PPA (60% glass fiber) are of the highest fracture toughness. These two elements perform quite well in the other categories and this can be seen in figures 26 and 27, due to their high fatigue strength and low thermal expansion coefficient.

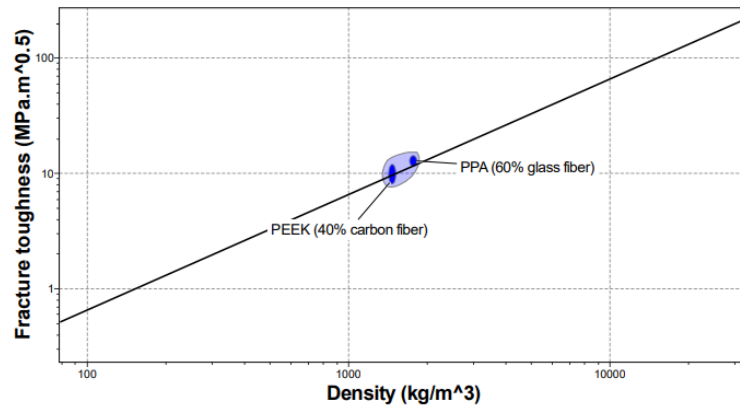


Figure 25. Ashby plot showing the top plastics for fracture toughness against density

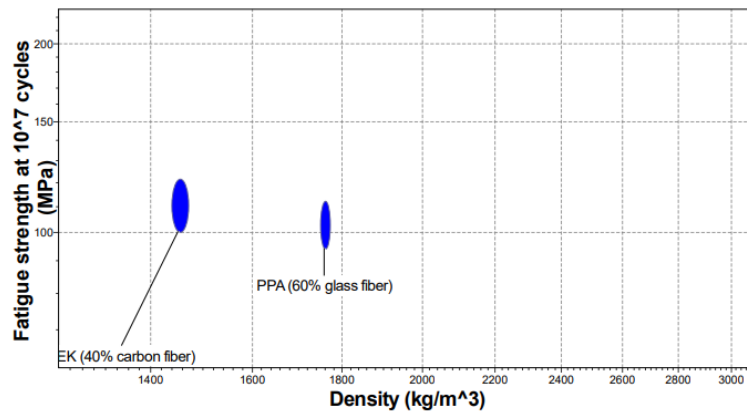


Figure 26. Ashby plot showing the top plastics for fatigue strength against density

Table 7 shows the comparison between the 2 plastics and the original choice of CFRP. Looking closely at the comparison between them, PEEK (40% carbon fibre) and PPA (60% glass fiber) are only slightly

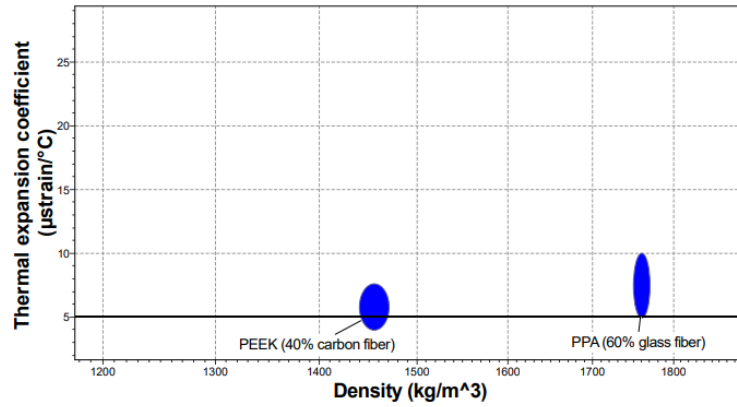


Figure 27. Ashby plot showing the top plastics for thermal expansion coefficient against density

less dense as CFRP, but PEEK (40% carbon fibre) is significantly more expensive. The 2 plastics have significantly less specific strength, specific stiffness and fracture toughness. Given this comparison, it was decided that the strength of CFRP is required and thus, its benefits outweighs the high cost.

Table 7. Comparing CFRP and the two plastic materials

	PEEK (40% carbon fibre)	PPA (60% glass fiber)	CFRP
Specific strength (kNm/kg)	165 - 185	144 - 146	700
Fatigue strength (MPa)	100 - 121	94.2 - 112	500 - 700
Specific stiffness (MNm/kg)	22.9 - 27.5	10.8 - 14.2	43.750
Fracture toughness (Mpa.m ^{0.5})	8.32 - 12	11.8 - 14.3	21.8750
Thermal expansion coeff. (10 ⁻⁶)	4 - 7.6	5-10	-0.1 - 1.7
Density (kg/m ³) (10 ³)	1.44 - 1.47	1.75 - 1.77	1.6
Cost (GBP/kg)	35.2 - 41.1	2.85 - 4.51	21.5